



Software Process Improvement –SEPI-471



Value Stream Mapping

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Agenda

- Value Stream Mapping

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Value Stream Mapping

Value Stream Mapping is a lean manufacturing technique to analyze, design, and manage the flow of materials and information required to bring a product to a customer.”

Value Stream Mapping

- Value Stream Mapping (VSM) is an excellent **tool for improving process and flow and for eliminating waste.**
- VSM is a method through which a team **maps out the flow of value for repeating processes**, forcing the team members to analyze where value is being added and where it is not.

Value Stream Mapping Components

- The value stream is the sequence of activities required to design, produce and deliver a good or service to a customer.
- **The value stream mapping parts**
 - Flows of information
 - Flow of material/Product Flow/Work Product Flow
 - Lead Time Ladder

Value Stream Mapping

- A team-based methodology for analyzing the current state and designing a future state for a series of events that take a product or service from its beginning through to the customer.

Value Stream Mapping

- Value stream mapping is a **lean management technique** that provides organizations with a visual representation of their end-to-end processes.
- It is a methodology **used to analyze, understand, and improve the flow of materials, information, and activities across the value chain.**
- **By examining the current state of operations and identifying areas of waste, value stream mapping enables businesses to identify opportunities for process optimization and continuous improvement.**

Value Stream Mapping

- Value stream mapping involves creating a detailed map or diagram that illustrates the entire **value stream from the moment a product or service is requested by a customer until it is delivered.**
- The map captures all the steps, activities, and interactions that occur along the way, including both value-added and non-value-added processes.

Value Stream Mapping

- A value stream map typically consists of various symbols and icons to represent different elements of the process.
- These symbols include boxes to represent processes or activities, arrows to indicate the flow of materials or information, and data boxes to capture key metrics such as cycle time or lead time.
- Other symbols such as triangles, clouds, or parallel lines can be used to denote inventory, delays, or decision points, respectively.

VSM and Software Development

- Value Stream Mapping is important in software engineering because it provides a structured approach to understanding and optimizing the entire software development process.
- In a complex field like software engineering, where collaboration, efficiency, and speed are paramount, VSM serves as a valuable tool to identify bottlenecks, reduce waste, and enhance productivity.

VSM and Software Development

- For example, consider a software development team that uses VSM to map their development process. They may discover that code reviews are taking a long time due to a backlog of code waiting for review.
- By visualizing this bottleneck, they can implement changes such as introducing automated code review tools or redistributing review responsibilities among team members. This leads to faster code reviews, shorter development cycles, and ultimately, quicker delivery of software to customers.

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VSM and Software Development

- Value Stream Mapping encourages a culture of continuous improvement within software engineering teams. By regularly reviewing and optimizing their value stream, teams can adapt to changing requirements and market demands more effectively.
- VSM empowers software engineering teams to be more efficient, agile, and responsive in a rapidly evolving industry.

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VSM Benefits

1. **Reducing Cycle Time** – you can decrease the time it takes to deliver a working piece of software by using stream mapping.
2. **Improving quality** – a value stream doesn't have to stay the same throughout the duration of the deployment process. Good stream management supported by measuring the most relevant metrics enables you to identify defects and root causes for delays and take action to prevent them from moving forward. This will ensure a better quality of your deployment processes and of the work needed to bring value to customers.
3. **Optimizing workflows** – value stream mapping can help you visualize the flow of work from idea to delivery by identifying bottlenecks and delays.

VSM-Concepts

- **Cycle time** (C/T) represents the amount of time needed to complete a task.
- **Setup Time** (S/T) or Change Over time is the necessary preparation time for an operation or process.
- **Uptime (%)** is the percentage of time a process or system is active.
- **Lead Time** sets the completion time for the entire operation/process.

VSM-Concepts

- **Takt Time** Frequency or pace of production required to meet customer demand. Defined as **available time divided by customer demand**.

Steps of VSM

Define the Scope: It's important to initially determine the boundaries and **scope of the value stream to map, identifying specific processes, departments, or areas that will be included in the map.**

Identify the Start and End Points: Clearly define the starting and ending points of the value stream. This could be the **initiation of a customer request or the delivery of a product or service.**

Gather Information: Collect data and information about the current state of the value stream. This can include process maps, flowcharts, metrics, and feedback from employees and customers.

Steps of VSM

Create a Value Stream Mapping Template: Utilize a value stream mapping template. These templates typically include symbols and icons to represent various elements of the value stream.

Draw the Current State Map: Using the template, start drawing the current state value stream map. Begin with the first step and connect each process with arrows to depict the flow.

Steps of VSM

Identify Value-Added and Non-Value-Added Activities: Analyze each step in the value stream to distinguish between value-added and non-value-added activities. Value-added activities directly contribute to meeting customer requirements, while non-value-added activities are wasteful and can be eliminated or minimized.

Identify Metrics and Data Points: Capture key metrics such as cycle time, lead time, and process efficiency in the data boxes. These metrics will help identify areas of improvement and serve as a basis for measuring future progress.

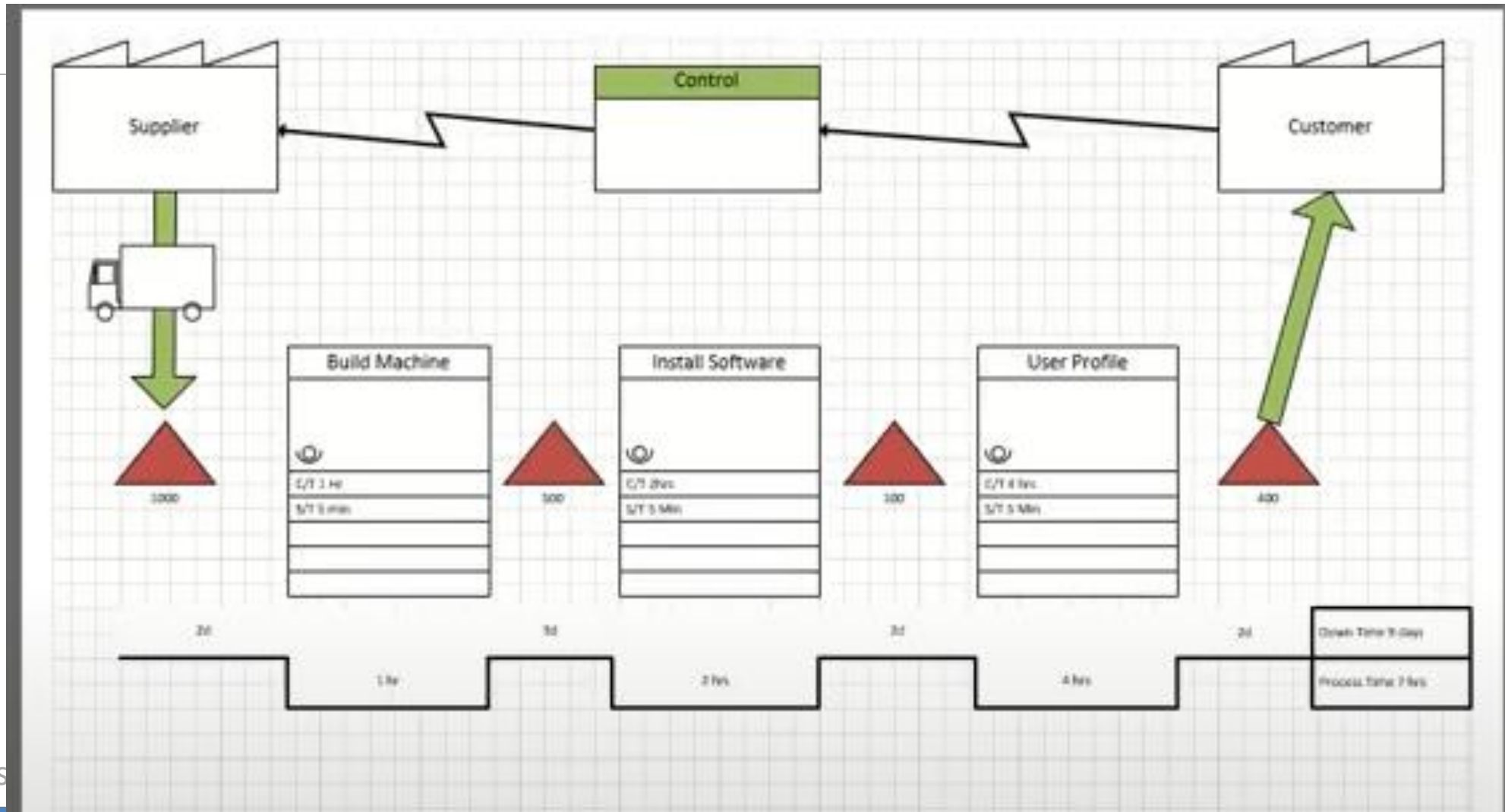
Steps of VSM

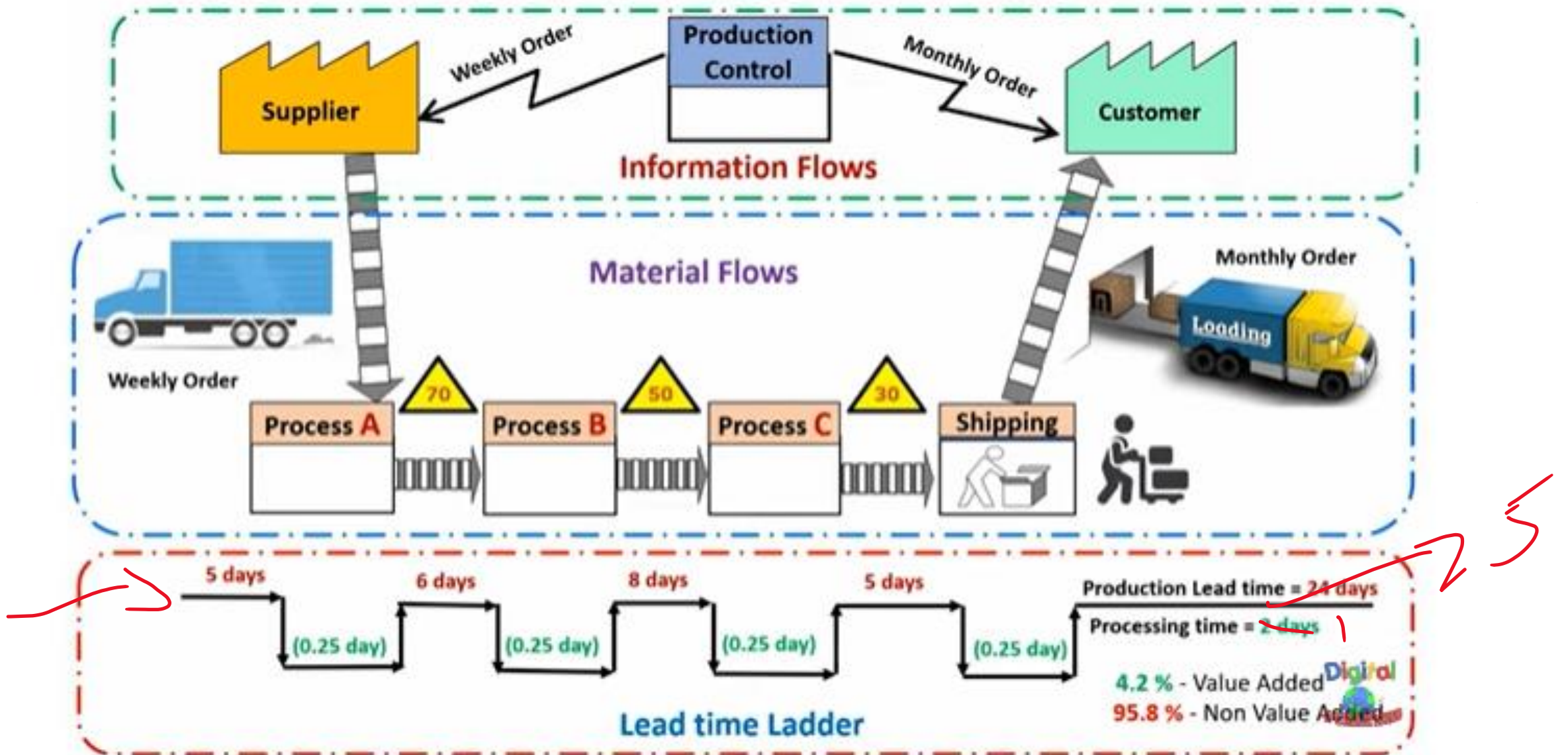
Analyze and Identify Improvement Opportunities: Review the current state map to identify bottlenecks, delays, excessive handoffs, or other areas of waste. Look for opportunities to streamline processes, reduce cycle times, and enhance overall efficiency.

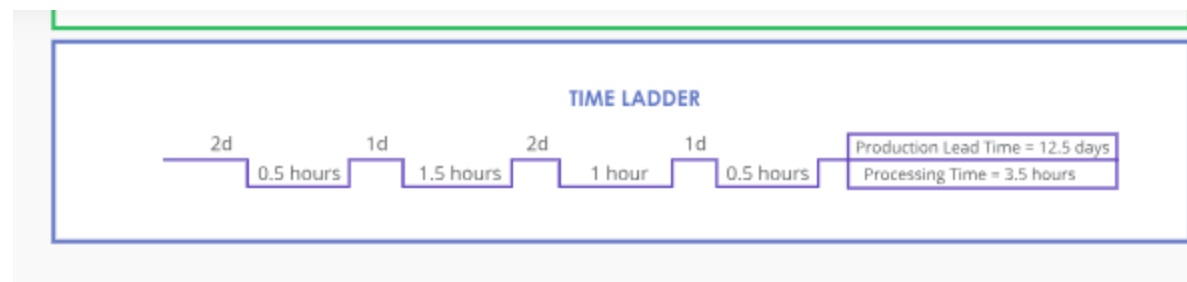
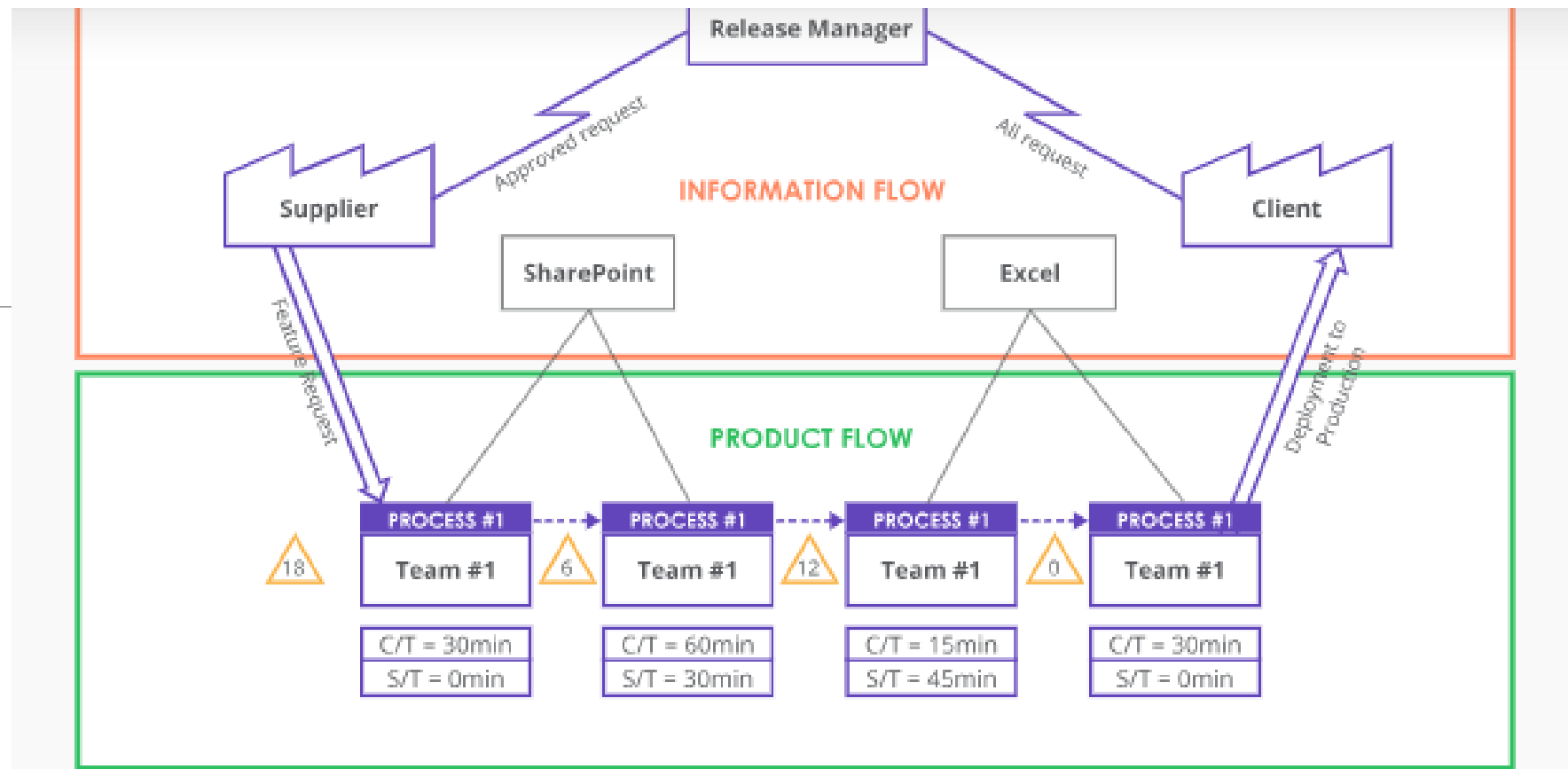
Design the Future State Map: Based on the analysis, envision the desired future state of the value stream. Redesign processes, eliminate waste, and optimize flow to create an improved value stream. Use the symbols and icons to draw the future state map.

Develop an Action Plan: Create a plan that outlines the steps and actions required to implement the changes identified in the future state map.

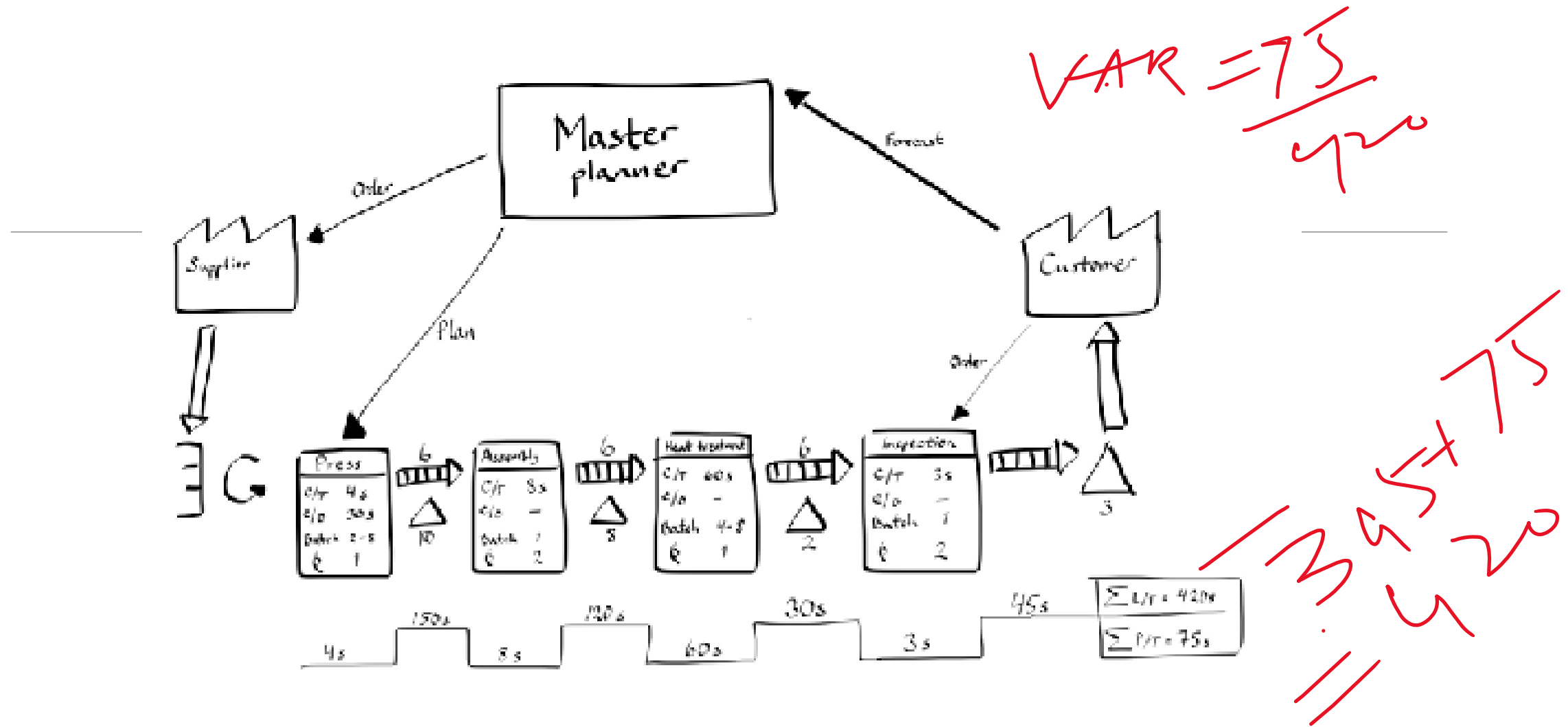
VSM



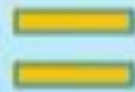




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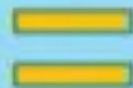


Value
Added
Ratio
(VAR)



The value added ratio (VAR) is the **time spent adding value** to a product or service, divided by **the total time** from the receipt of an order to its delivery.

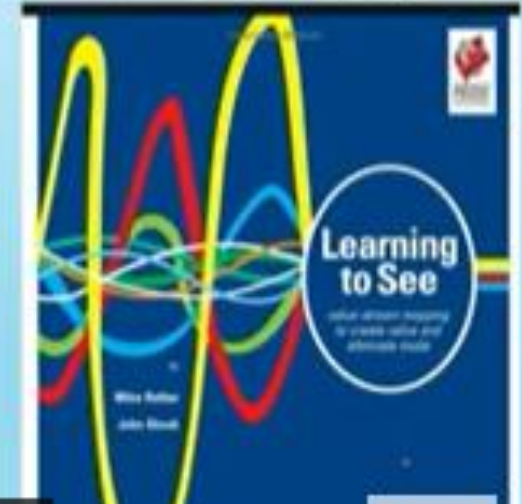
Value Added
Ratio
(VAR)



186 sec

23.6 Days – 2039040 sec

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Thank you

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